

Transparent policy evaluation through causal machine learning

An application to SME digital subsidies

Eugeni Gil-Ocana Ana Garcia-Bernabeu Pablo de Pedraza

Universitat Politècnica de València

Data for Policy 2026



eSMART

- 1. Motivation and objectives** *what do we do?*
- 2. The framework** *how do we solve it?*
- 3. The Kit Digital programme** *with which data?*
- 4. Results and validation** *what do we find?*
- 5. Conclusions and limitations** *what to take home?*

Part 1 of 5

Objectives and motivation

1. Objectives and motivation
2. The framework
3. The Kit Digital programme
4. Results and validation
5. Conclusions and limitations

Purpose and objectives

We evaluate the impact of a public subsidy with a **causal random forest**, and make the evaluation **auditable**.

Applied

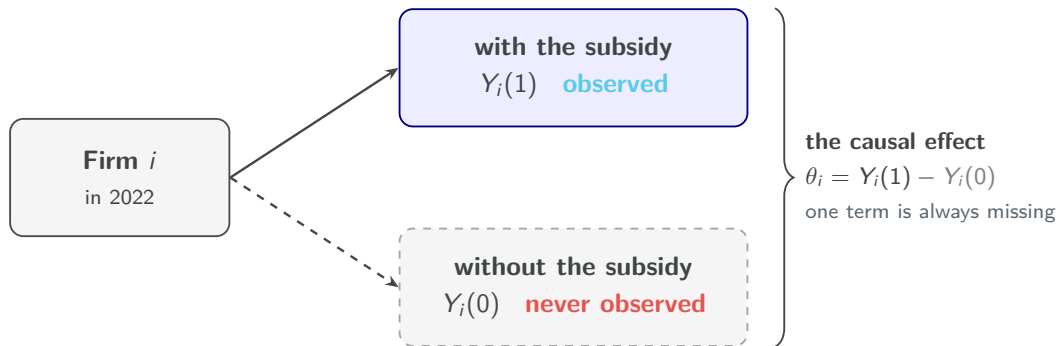
Estimate the effect of the voucher on the revenues, value added and employment of SMEs, one and two years after treatment.

Methodological

Algorithmic transparency: the black-box estimator explained (SHAP) and summarised with valid inference (BLP)

Data: two administrative registries (BDNS + SABI) linked into a reusable firm-level evaluation panel

Causal evaluation is a missing-data problem



"The fundamental problem of causal inference" — Holland (1986)

Evaluating a policy = reconstructing the missing outcome from **comparable untreated firms**

Conventional parametric heterogeneity vs flexible causal ML

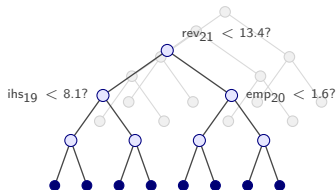
Traditional econometrics

- **One average effect** for the whole sample
- **Functional form fixed in advance**: how characteristics enter the model
- **Heterogeneity only if pre-specified**: interactions chosen before seeing the data
- **Transparent by construction**: coefficients, standard errors

Causal machine learning

- **An effect for every firm**: the expected effect for a firm with profile x
- **Relationships learned from the data**: no functional form imposed
- **Heterogeneity discovered**, not pre-specified: who responds is part of the answer
- **Not transparent**: results difficult to interpret

The transparency gap



× 2,000, averaged

an effect for every firm,
no account of why

- 1 No explicit rule: nothing to report, examine, or contest.

→ we **explain it**: every estimate attributed to the firm's characteristics

- 2 No readable summary: 65,131 effects, but no coefficients or standard errors.

→ we **summarise it**: a linear projection with valid causal inference

- 3 No visible limits: opacity hides how fragile the conclusions are.

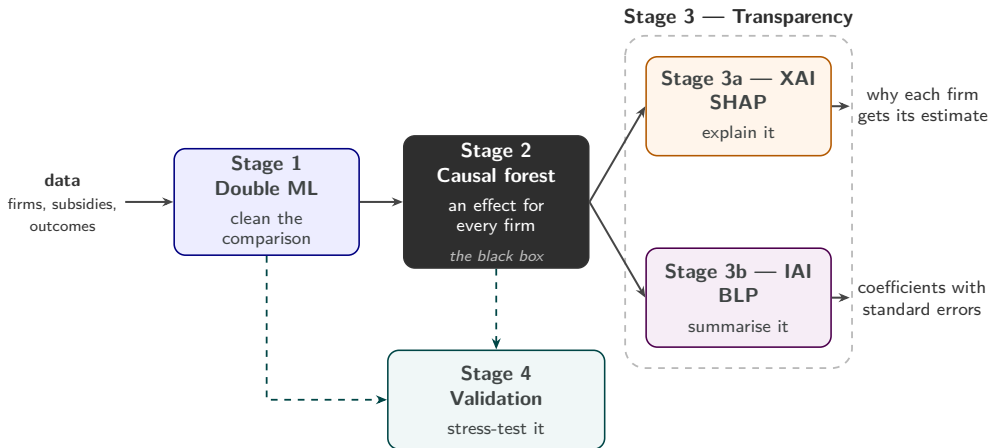
→ we **test it**: overlap, pre-trends, and how much confounding it would take

Part 2 of 5

The framework

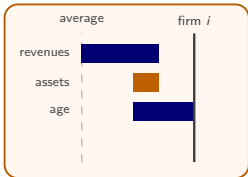
1. Objectives and motivation
2. The framework
3. The Kit Digital programme
4. Results and validation
5. Conclusions and limitations

An auditable evaluation workflow



Three devices, three questions

XAI — SHAP



which characteristics led the model to assign this firm its effect?

IAI — BLP

slope of the effect	
size	−0.020***
revenues	−0.013
firm age	−0.005

is the pattern distinguishable from zero, and in which direction?

Validation

- ✓ overlap
- ✓ pre-trends
- ✓ sensitivity

how robust are the results?

Part 3 of 5

The Kit Digital programme

1. Objectives and motivation
2. The framework
3. The Kit Digital programme
4. Results and validation
5. Conclusions and limitations

The Kit Digital programme

Why it matters

- SMEs are **99.8%** of Spanish firms (Ministerio de Industria, DIRCE)
- The digital gap: **73%** of SMEs vs **98%** of large firms reach basic digital intensity in the EU (Eurostat, 2025)
- **€3.07 bn** from Next Generation EU, managed by Red.es — **200,000+** firms by 2024, one of the largest digitalisation programmes in Europe

The programme

- A **voucher**, redeemable only through accredited providers: websites, e-commerce, cybersecurity, e-invoicing
- Capped by firm size: €2k (<3 employees), €6k (3–9), €12k (10–49)

Data and variables

BDNS subsidy registry + **SABI** firm accounts → firm-level panel, 2019–2024

65,131 firms: 13,530 treated in 2022, 51,601 never treated (2021–2023)

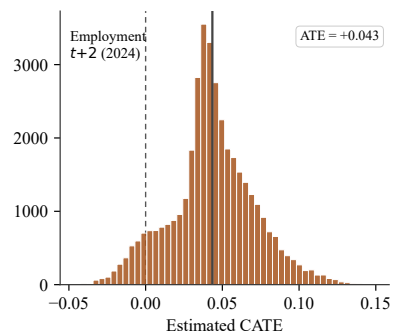
Y	Outcomes	revenues, value added, employment (logs), one and two years after treatment
T	Treatment	first Kit Digital disbursement in 2022; controls never enter the programme
X	Pre-treatment profile	2019–2021 trajectories of the three outcomes, plus assets, firm age, sector (NACE) and region (NUTS-1)

Part 4 of 5

Results and validation

1. Objectives and motivation
2. The framework
3. The Kit Digital programme
4. Results and validation
5. Conclusions and limitations

Average and conditional effects



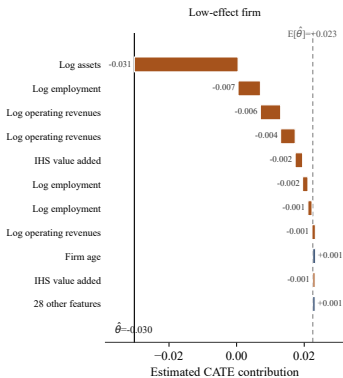
92.9% of firms with a positive estimated effect
employment, two years after treatment

Outcome		ATE	<i>p</i>
Revenues	<i>t</i> +1	+0.023	0.27
	<i>t</i> +2	+0.028	0.21
Value added	<i>t</i> +1	+0.019	0.36
	<i>t</i> +2	+0.029	0.25
Employment	<i>t</i> +1	+0.026	0.08
	<i>t</i> +2	+0.043	0.01

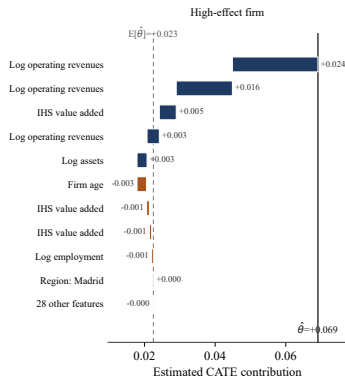
causal-forest ATEs, log points $\approx$ %

Transparency: why each firm gets its estimate

start at the average effect $\rightarrow$ add one characteristic at a time $\rightarrow$ land exactly on **the firm's own estimate**



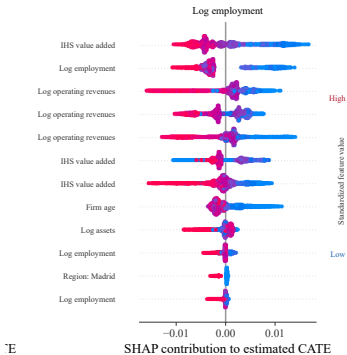
a **large** firm — scale pulls it **below** the average



a **small** firm — scale pushes it **above** the average

Transparency: what drives the heterogeneity

each dot is a firm; right of zero **raises** its effect, left **lowers** it; **red** = high value, **blue** = low



employment, two years after treatment; features ordered by importance, repeated names are the 2019–2021 values

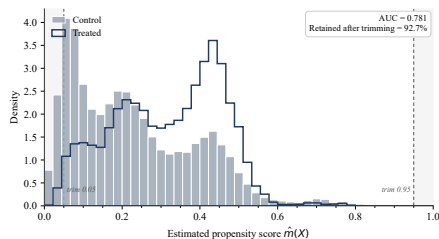
Interpretability: the summary a reader can contest

	Employment $t+1$	Employment $t+2$
How large is the effect? group average effects by size segment		
Micro (<3 employees)	+0.050**	+0.072***
Small (3–9)	+0.039**	+0.059***
Medium (10–49)	+0.015	+0.029**
How does it change with baseline traits? slopes per SD — <i>not</i> effects		
Log employment (2021)	−0.017***	−0.020***
Log operating revenues (2021)	−0.009	−0.013
IHS value added (2021)	−0.003	+0.005
Log assets (2021)	+0.002	−0.001

negative slope = smaller firms gain more; every effect above is positive. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

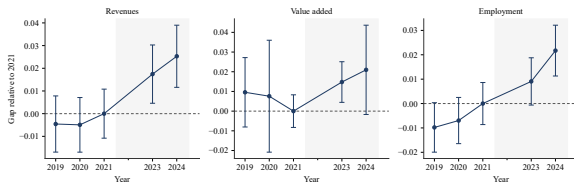
Validation and robustness

Overlap



AUC 0.781, far from perfect separation —
comparable firms across the whole range

Pre-treatment trends



gaps flat in 2019–2020 for all three outcomes —
indistinguishable paths through the pandemic

Sensitivity: overturning **employment** needs a hidden confounder $\sim 6\times$ stronger than anything observed. Revenues and value added do not survive.

Part 5 of 5

Conclusions and limitations

1. Objectives and motivation
2. The framework
3. The Kit Digital programme
4. Results and validation
5. Conclusions and limitations

Conclusions

An evaluation that is transparent and auditable,
is more useful than one that reports only estimates.

explained

SHAP

summarised

BLP, with inference

stress-tested

validation layer

Takeaways

The case study

- Employment **+4.3%** at two years — the finding that **survives** scrutiny
- Concentrated among **micro and small** firms, driven by pre-treatment scale
- Revenues and value added: same sign, fragile to modest unobserved confounding

The framework

- A **black-box causal estimator can be made auditable**: explained, summarised with inference, stress-tested
- Transfers to any scheme with a beneficiary register, outcome microdata and **rule-based eligibility**
- Open-source tooling, **replication code public**

Limitations

- **Identification, not estimation, binds.** Transparency does not remove unobserved confounding it quantifies how much would overturn each result.
- **Two years may be short.** Digitalisation may not reach revenues within the window; employment may respond first.

References

- Holland (1986). Statistics and causal inference. *JASA*.
- Chernozhukov et al. (2018). Double/debiased machine learning. *Econometrics J*.
- Wager and Athey (2018); Athey and Wager (2019). Causal forests. *JASA*; *Obs. Studies*.
- Lundberg and Lee (2017). A unified approach to interpreting model predictions (SHAP). *NeurIPS*.
- Semenova and Chernozhukov (2021). Best linear projection. *Econometrics J*.
- Chernozhukov et al. (2022). Long story short: omitted-variable bias in ML. *NBER*.
- Rehill and Biddle (2024). Transparency challenges in policy evaluation with causal ML. *Data & Policy*.
- Amarasinghe et al. (2023). Explainable ML for public policy. *Data & Policy*.
- Red.es (2022). Programa Kit Digital.
- Eurostat (2025). Digitalisation in Europe.

Thank you

Transparent policy evaluation through causal machine learning

An application to SME digital subsidies

Eugeni Gil-Ocana | Ana Garcia-Bernabeu | Pablo de Pedraza

Universitat Politècnica de València

Full paper under review at *Data & Policy*, DAP-2026-0220

egiloca@doctor.upv.es | replication code: github.com/EugeniGil/research-code